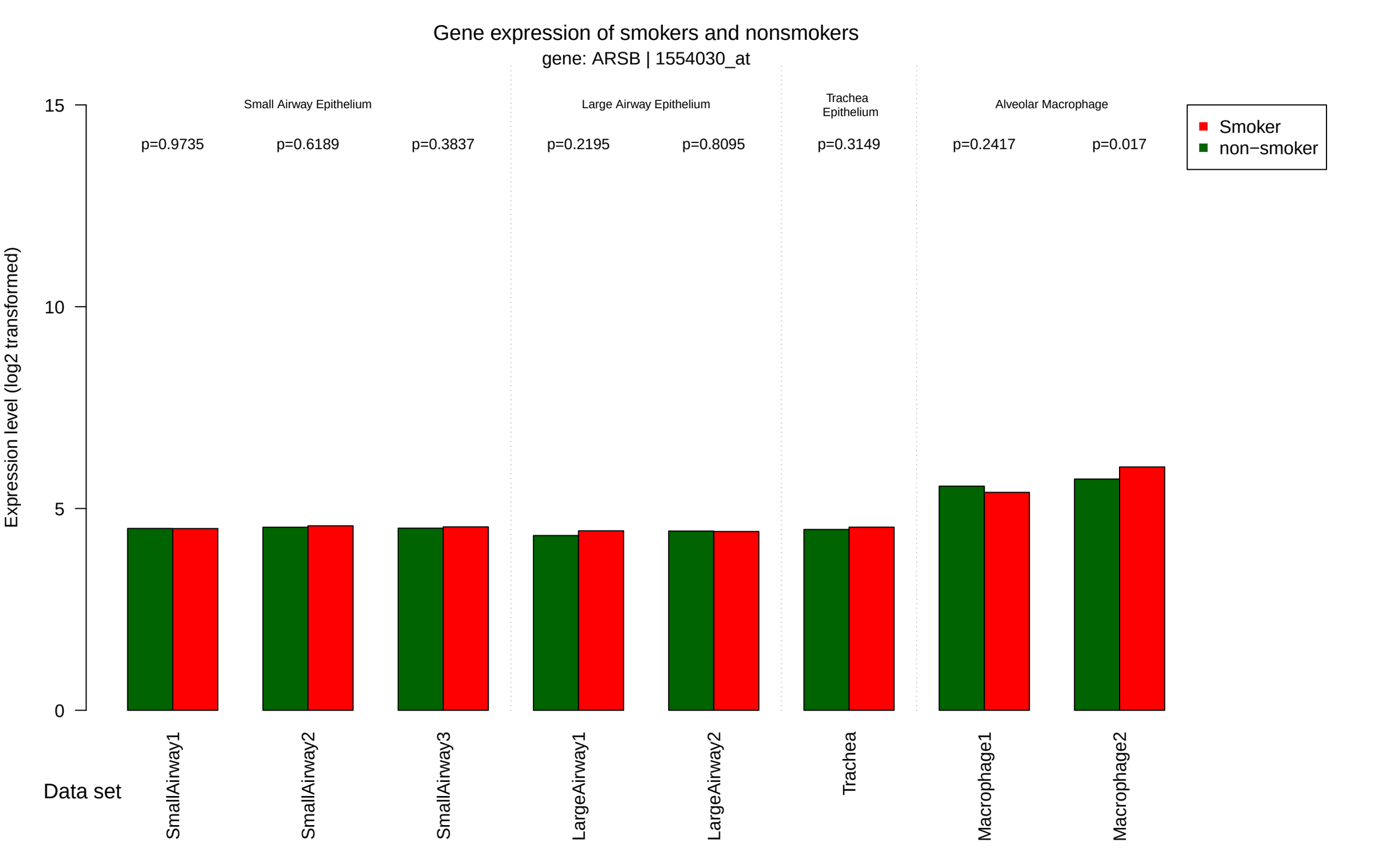
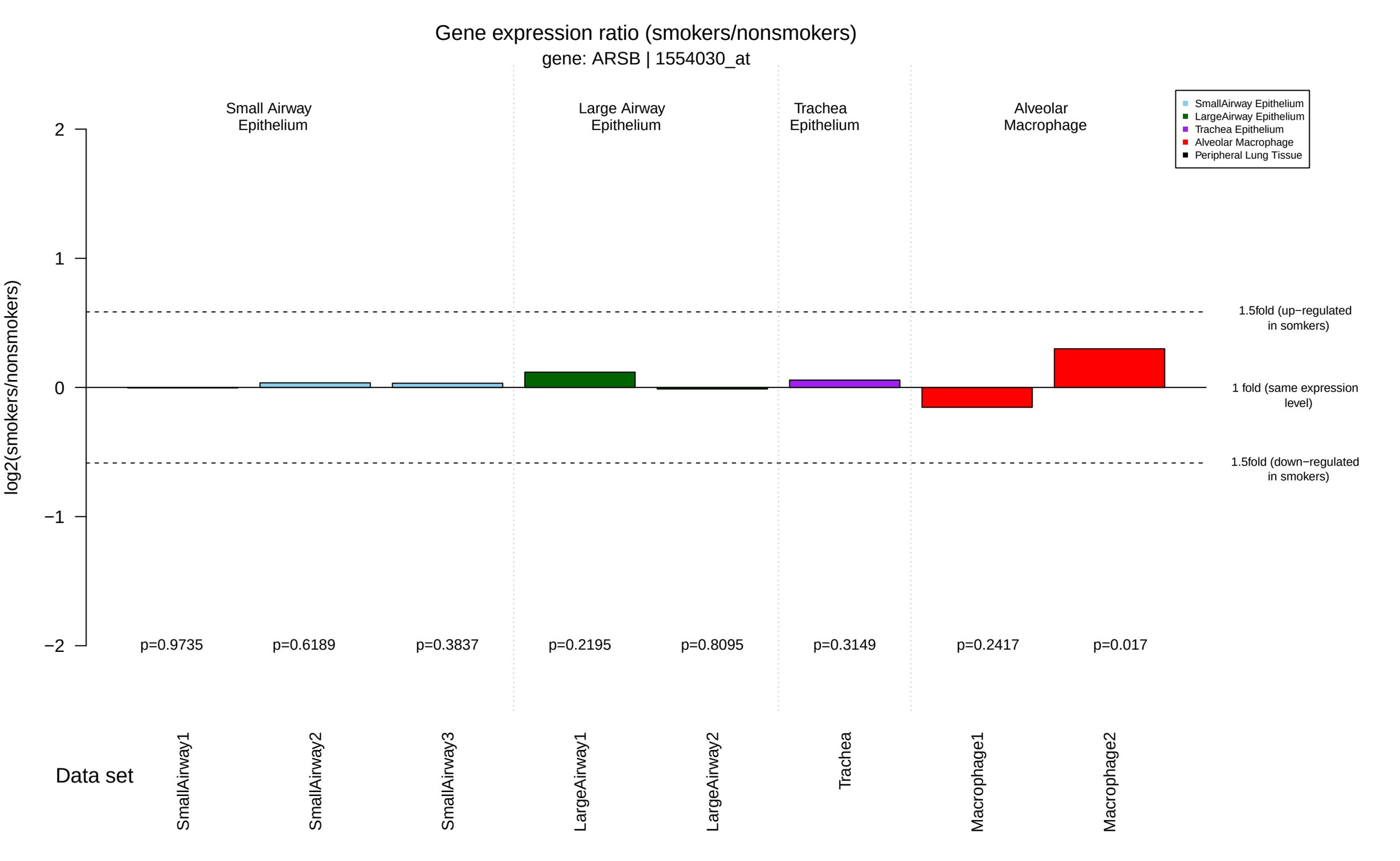
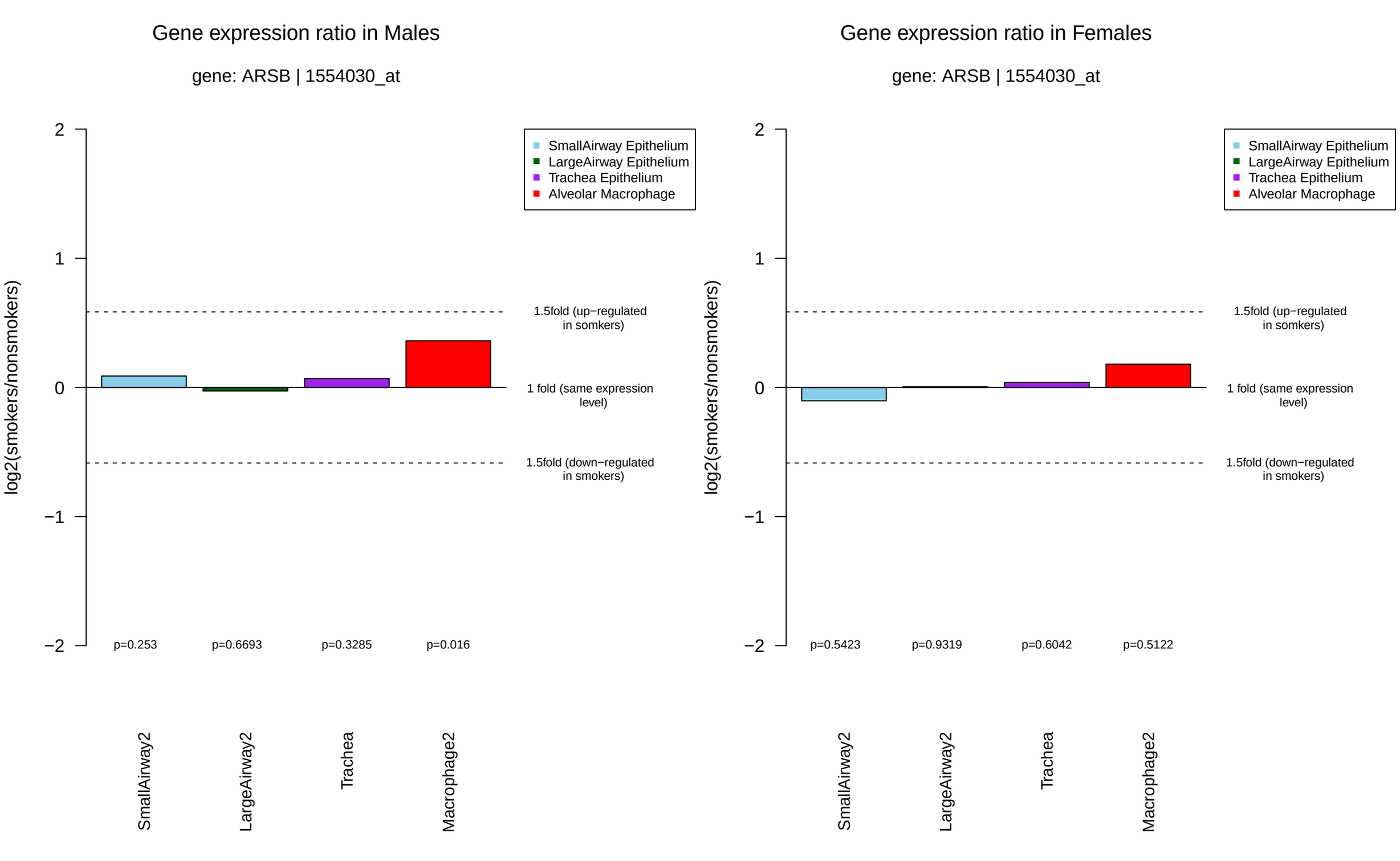
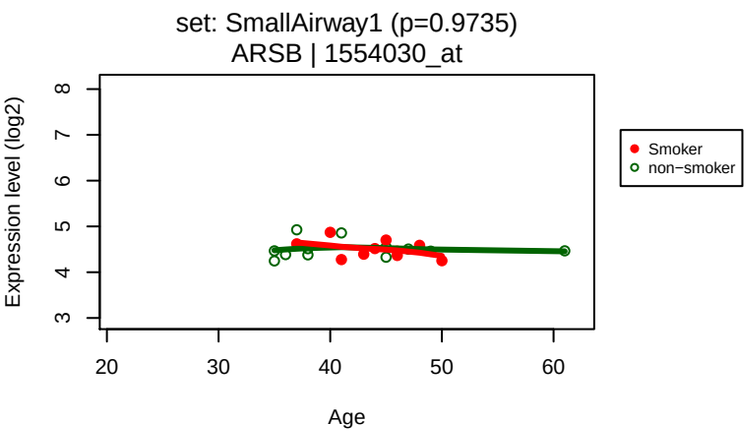
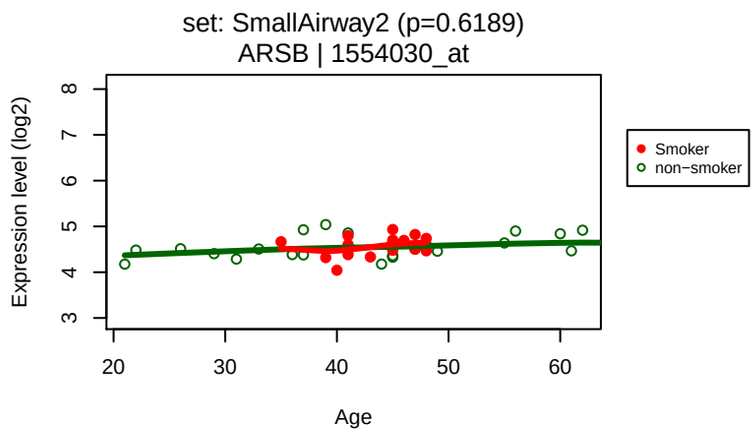
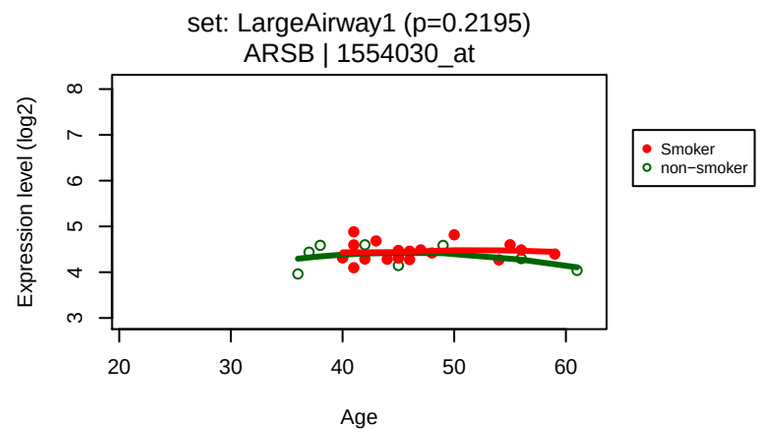

Gene expression of smokers and nonsmokers
gene: ARSB | 1554030_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9735
p=0.6189
p=0.3837
p=0.2195
p=0.8095
p=0.3149
p=0.2417
p=0.017
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

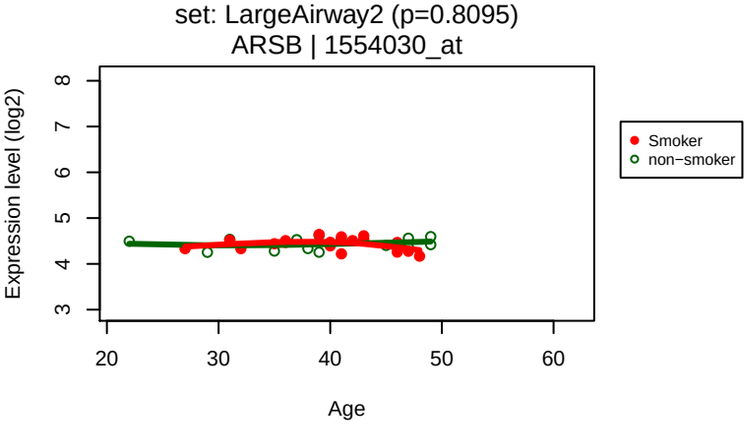

Gene expression ratio (smokers/nonsmokers)
gene: ARSB | 1554030_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9735
p=0.6189
p=0.3837
p=0.2195
p=0.8095
p=0.3149
p=0.2417
p=0.017
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

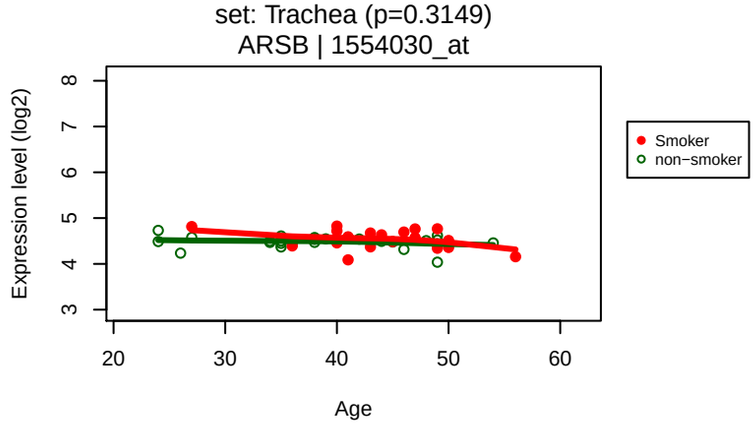

Gene expression ratio in Males
gene: ARSB | 1554030_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.253
p=0.6693
p=0.3285
p=0.016
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARSB | 1554030_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5423
p=0.9319
p=0.6042
p=0.5122
SmallAirway2
LargeAirway2
Trachea
Macrophage2

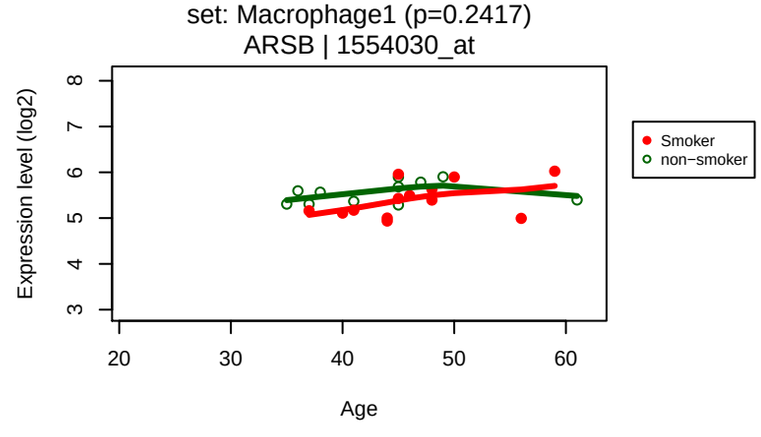

set: SmallAirway1 (p=0.9735)
ARSB | 1554030_at
Expression level (log2)
Age
Smoker
non−smoker

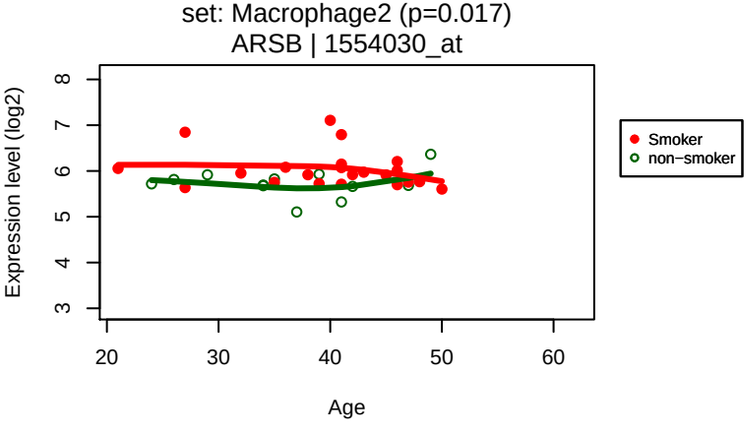

set: SmallAirway2 (p=0.6189)
ARSB | 1554030_at
Expression level (log2)
Age
Smoker
non−smoker

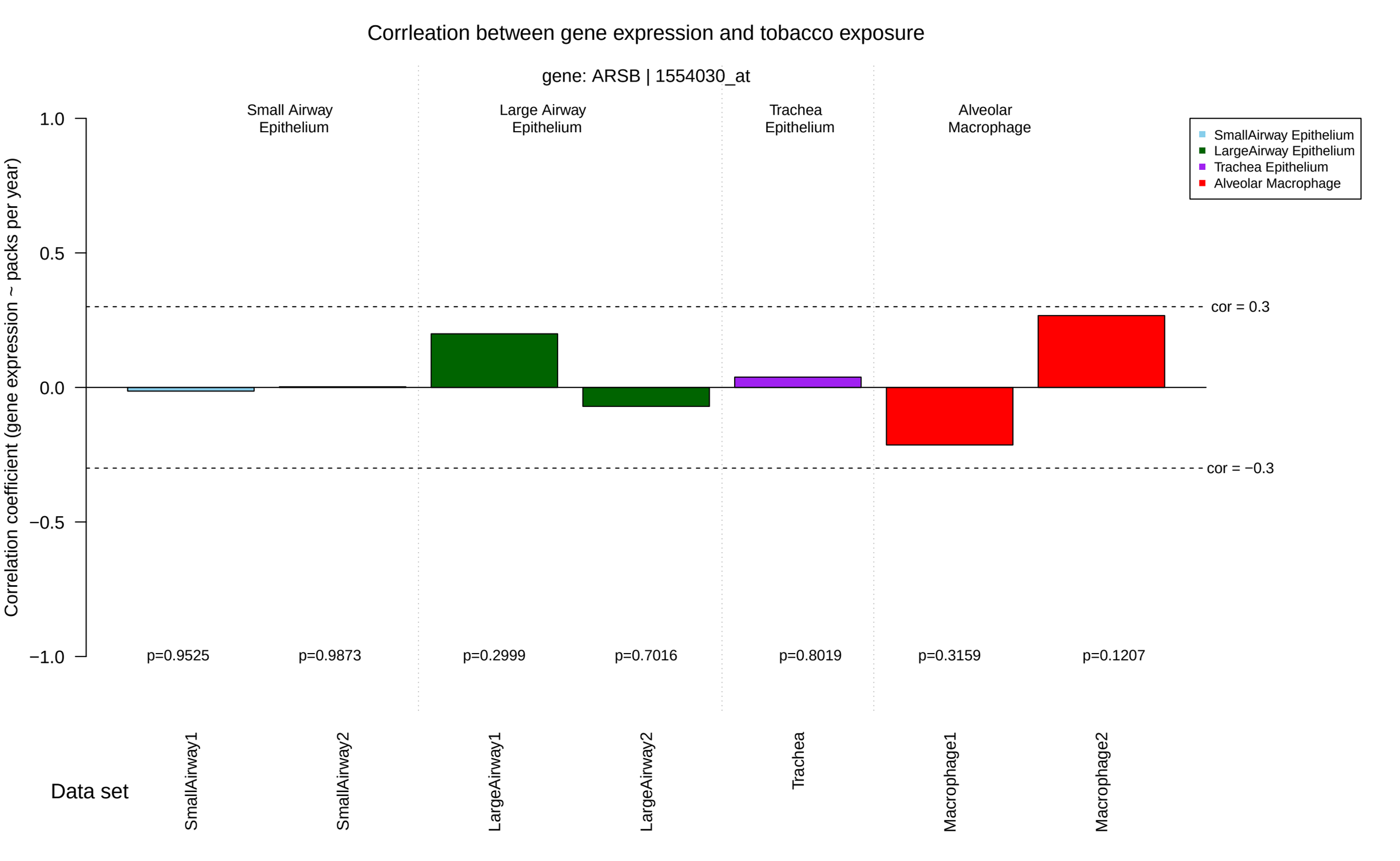

set: LargeAirway1 (p=0.2195)
ARSB | 1554030_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8095)
ARSB | 1554030_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3149)
ARSB | 1554030_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2417)
ARSB | 1554030_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.017)
ARSB | 1554030_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARSB | 1554030_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9525
p=0.9873
p=0.2999
p=0.7016
p=0.8019
p=0.3159
p=0.1207
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2